

Crowded Carry in G10 FX and Crypto

Crash risk in FX and crypto carry trades

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Carry pays because it crashes

Same economic mechanism, two very different markets

- Carry: borrow the low-yield asset, hold the high-yield one.
- UIP says this earns nothing; in the data it earns a premium, with rare large crashes.
- Crashes are worst when the trade is crowded and everyone unwinds at once (Brunnermeier, Nagel & Pedersen 2008).

Hypotheses, fixed before testing:

- A: carry earns a premium; crowded months are followed by weaker carry; it loses in volatility spikes.
- C: funding carry earns a premium with little BTC exposure; its crashes are coin-specific.

A: G10 FX

C: Crypto

Carry signal	3-month rate differential	Perpetual funding rate
Crowding data	CFTC speculator positions	Funding = leveraged demand
Universe	7 currencies vs USD	12 coins incl. LUNA, FTT
Frequency	Monthly, 1999–2026	Weekly, 2020–2026

Data and methodology

Public data; a research design chosen to keep the tests honest

Data	Source	Coverage
FX spot, 7 currencies	FRED (Fed H.10)	1999–2026
3-month rates	OECD via FRED	1999–2026
Speculator positions	CFTC Commitments of Traders	1986–2026
Crypto price, volume, funding	Binance public archive	2020–2026
Carry factor (validation)	Lustig–Roussanov–Verdelhan	1983–2021

Data fix: before Aug 2000 CFTC lists the same currency futures under 'International Monetary Market'. Matching both names extended positioning back to 1986.

- Parameters fixed before the first backtest
- Out-of-sample: A from 2011, C from 2024
- Costs: 3 bp per unit traded in FX, 5 bp in crypto
- No look-ahead: weights at t earn $t+1$; CFTC used after its Friday release
- No survivorship bias: LUNA and FTT kept
- Robustness grids reported, never used for tuning

Strategy A: crowding-filtered G10 carry

JPY, EUR, GBP, CHF, CAD, AUD, NZD vs USD, monthly

1

Carry portfolio

Each month-end: long the 2 highest-rate currencies, short the 2 lowest. Equal weights, dollar-neutral.

2

Crowding signal

CFTC net speculative position / open interest, as a 36-month z-score.

3

Portfolio crowding

Average z of the long leg minus average z of the short leg.

4

Filter

If crowding > 1, cut all positions to half for the next month.

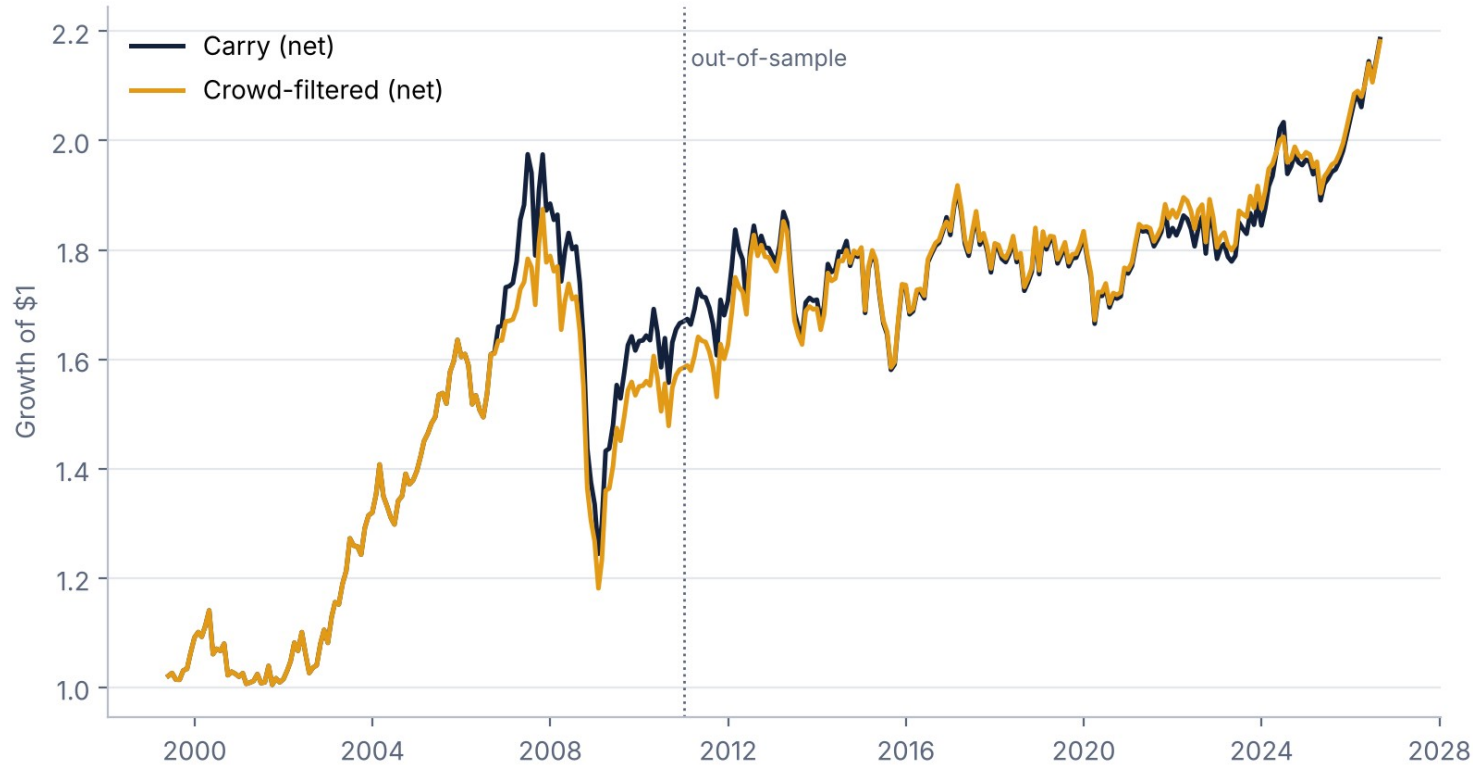
33

crowded months out of 328, 1999–2026

1 SD

Threshold for 'crowded': one standard deviation above normal positioning

Crowding filter helps out of sample, not in 2008



Full sample, net: Sharpe 0.37 carry, 0.38 filtered; max drawdown -37% either way.

0.27 → 0.32

Out-of-sample Sharpe, net: carry vs crowd-filtered (2011–2026)

0.30% vs 0.02%

Next-month carry after non-crowded vs crowded months

0.82

Correlation with the academic LRV carry factor (1999–2021)

Strategy A loses when volatility spikes

Monthly net carry on risk factors, Newey-West t-stats, 1999–2021

-1.55

Beta to equity-volatility changes (t = -3.7)

0.35

Beta to the dollar factor (t = 4.1)

0.25%

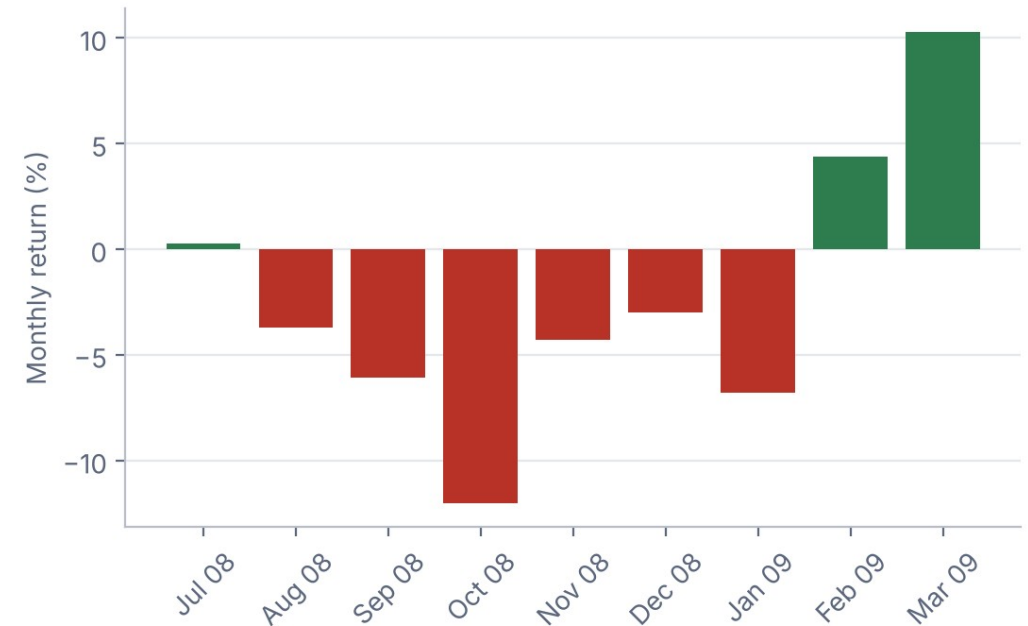
Monthly alpha (t = 1.7): not significant

0.18

R² of the two-factor model

Worst months: Oct 2008, Aug 2007, May 2000, Jan 2009, and the Jan 2015 Swiss franc de-peg.

The 2008 carry crash: -20.5% (Jul 2008 – Mar 2009)



Strategy C: crypto funding carry

12 Binance perpetuals incl. LUNA and FTT, weekly, 2020–2026



- Funding is paid every 8 hours; positive funding means longs pay shorts.
- Each Sunday: short the third of coins with the highest 7-day funding, long the lowest third.
- Dollar-neutral, 5 bp costs.

0.14

Net Sharpe, full

0.81

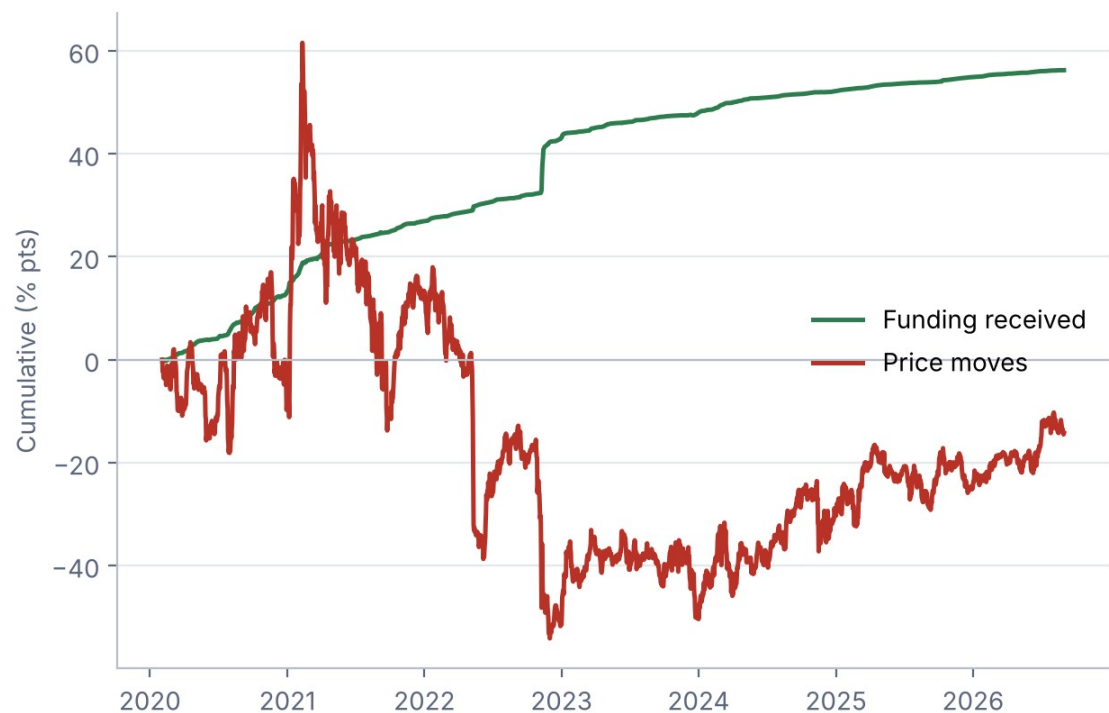
Net Sharpe, 2024–26

-27%

Worst week: LUNA collapse, May 2022

Funding is earned, but price moves dominate

Where the return comes from, and what drives the losses

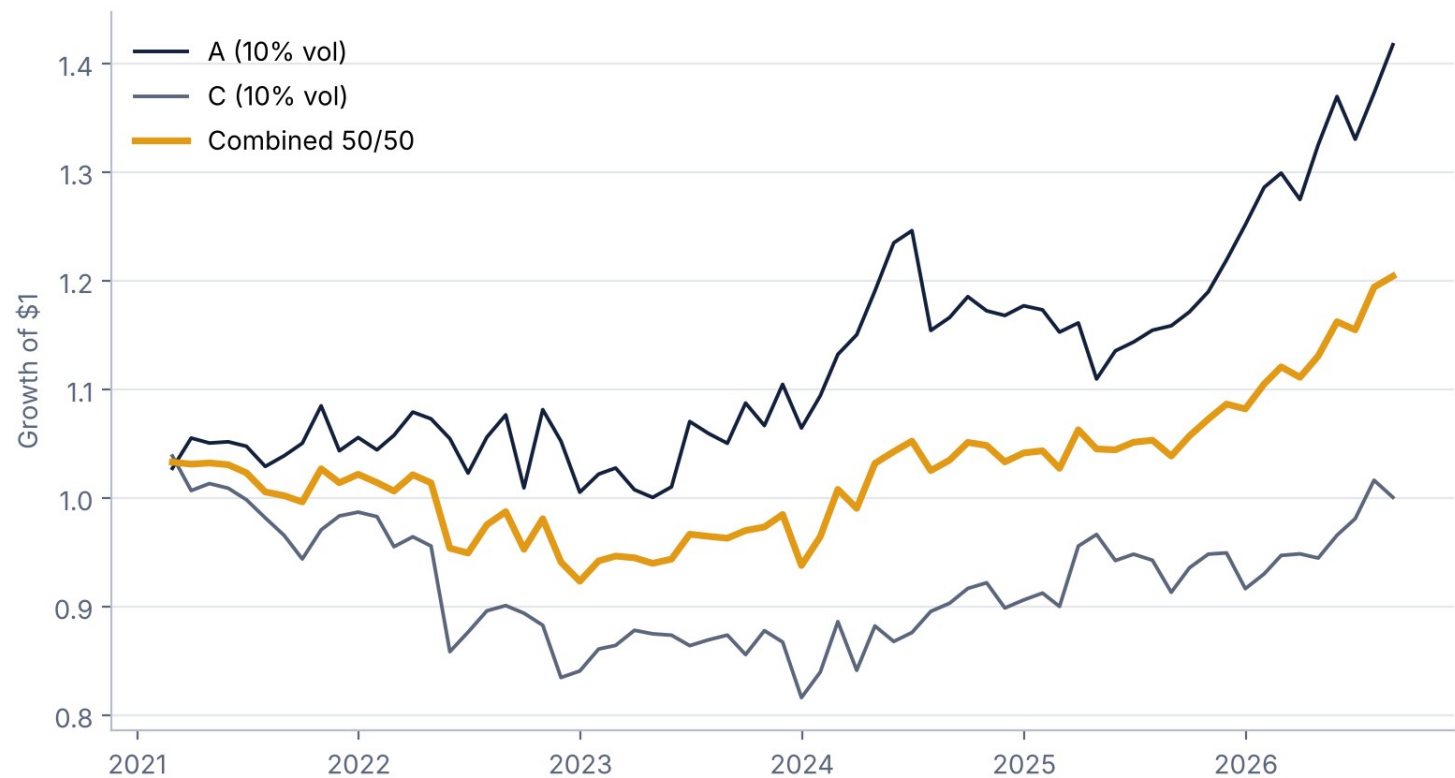


% per year	Funding	Price	Net
Full	+8.5	-2.2	+4.1
In-sample	+12.3	-12.5	-2.5
Out-of-sample	+3.1	+13.0	+13.9

- Beta to BTC ≈ -0.02 ($t = -0.75$): market-neutral.
- Losses are coin-specific: LUNA -87, XRP -58, DOGE -43 % pts.
- Pre-set risk control (1/6 cap, no longs below -50% funding): Sharpe 0.14 $\rightarrow$ -0.16. It did not help.
- **Lesson: deeply negative funding means crowded shorts, not distress.**

Combining FX and crypto carry

Each scaled to 10% vol with lagged volatility, 50/50, 2021-2026



0.11

Correlation between A and C

	Sharpe	Vol (%)	MaxDD (%)
A	0.71	9.4	-11
C	0.05	9.3	-21
Combined 50/50	0.51	7.0	-11

Conclusions and limitations

What we found

- FX carry reproduces the academic factor (corr 0.82) and is exposed to volatility shocks.
- Crowding predicts weaker carry and modestly improves the out-of-sample Sharpe.
- Crypto funding carry is market-neutral, but its crashes are coin-specific: it ends up long what traders short hardest.
- FX and crypto carry are nearly uncorrelated (0.11).

Limitations

- Short crypto history (2020–2026) and only 12 coins.
- Few crowded FX months (33); the 2008 crash was not flagged.
- Crypto results depend on lookback and rebalancing frequency.
- Costs are simple estimates; no funding-rate caps, borrow limits or exchange risk.